Parth Sachin Deshmukh

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ML Engineer building LLM fine-tuning pipelines, RAG systems, and computer vision models; First Author Springer 2024. Seeking full-time **ML Engineer / Applied Scientist** roles starting **May 2027**.

Education

Texas A&M University — MS, Computer Science — GPA: **4.0 / 4.0** *Courses: AI, NLP, Data Mining, Algorithms* Aug 2025 – May 2027

Vishwakarma Institute of Information Technology — BS, Information Technology — GPA: **9.12 / 10.00** Aug 2020 – May 2024

Skills

Languages: Python, C++, SQL, JavaScript, TypeScript

AI / ML: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, LangGraph, LLM Fine-Tuning (LoRA, QLoRA, PEFT), RAG, Multi-Agent Systems, Computer Vision, NLP, Scikit-learn, Prompt Engineering

LLMOps & Cloud: Docker, AWS (EC2, S3, SageMaker), GitHub Actions, CI/CD, FastAPI, LangSmith, Weights & Biases, Gradio, pytest

Data & Databases: ChromaDB (Vector DB), MySQL, BigQuery, Pandas, NumPy, OpenCV

Experience

ML / Data Research Intern — Indian Institute of Technology (IIT) Bombay *(Asia Top-5 Engineering School)* Feb 2024 – Sep 2024

- Deployed a full-stack analytics platform (Yii, MySQL) to **20+ faculty end-users**; gathered requirements, iterated on feedback, and trained stakeholders — cutting dashboard setup time by **~60%** via a Figma-to-HTML parser powering **100+ templates**.
- Optimized MySQL queries via indexing across **500+ configurations**; implemented AJAX async loading, improving UI response time by **~40%**.
- ***Skills:*** *PHP, JavaScript, TypeScript, Yii Framework, MySQL, AJAX, Stakeholder Communication*

Projects

LLM Fine-Tuning: Code Generation *GitHub* Apr 2026

- Fine-tuned **LLaMA 3.2 (3B)** on CodeAlpaca-20k using LoRA + 4-bit QLoRA (PEFT), cutting trainable parameters by **99%** and achieving **+65% ROUGE-L improvement** over the base model.
- Deployed adapter weights and a live **Gradio inference demo** to HuggingFace Hub with automated ROUGE-L & BERTScore evaluation.
- ***Skills:*** *PyTorch, LoRA, QLoRA, PEFT, HuggingFace Transformers, Gradio*

RAG-Powered Code Assistant *GitHub* Mar 2026

- Built a production-grade RAG pipeline (LangChain + LLaMA 3) enabling semantic search across **5,000+ documentation pages** with **sub-second latency** via ChromaDB vector store.
- End-to-end MLOps: published to **PyPI/Conda**, containerized with Docker, CI/CD via GitHub Actions with **85% test coverage** on Linux & Windows.
- ***Skills:*** *LangChain, ChromaDB, Docker, GitHub Actions, FastAPI, pytest, PyPI*

Multi-Agent LLM Research Assistant *GitHub* Jan 2026

- Designed a multi-agent system (LangGraph) with retriever, summarizer, and critic agents for autonomous multi-hop queries over a **10,000-document corpus**.
- Deployed as a FastAPI microservice; reduced query latency by **40%** vs. single-agent baseline via parallel agent execution.
- ***Skills:*** *LangGraph, OpenAI API, ChromaDB, FastAPI, Python*

Satellite Image Segmentation Pipeline *GitHub* Oct 2025 – Feb 2026

- Built a semantic segmentation pipeline using SAM2 to extract structural features from satellite imagery; achieved **mIoU 0.62** and **pixel accuracy 0.78** across 17 test images with GPU-optimized CUDA inference.
- ***Skills:*** *PyTorch, SAM2, CUDA, NumPy, OpenCV*

Publications & Achievements

- **First Author**, Springer 2024: *Automated GUI Testing to Enhance UX: A Survey of the State of the Art*
- **Co-Author**, IEEE 2024: *Increasing Donation Transparency in Disaster Relief: A Blockchain-based Solution*
- **Exchange:** Advanced Engineering & Data Science School, Tokushima University, Japan (2023–2024) **Cert:** AI Agents Fundamentals